

Rohan Bahl

rohan_bahl@u.nus.edu | rb9823.github.io/website/ | linkedin.com/in/rohan-bahl | github.com/RB9823 | +65 91471451

EDUCATION

National University of Singapore
Bachelor of Computing (Hons), Computer Science (Specialisations: Artificial Intelligence & Database Systems)
• Special Programme: **NUS Overseas Colleges Toronto–Waterloo AY 24/25 Sem 2 Intake**

Expected: Dec 2025

EXPERIENCE

Tudor Capital
Site Reliability Engineer Intern
• Achieved sub-second latency in managing high-volume Grafana alerts by designing a distributed alerting platform (4 Spring Boot microservices with ActiveMQ + Redis), replacing ad-hoc workflows
• Reduced Grafana infrastructure management time by 99% (10 hrs → 5 mins) by building an Ansible + Python automation framework leveraging the Grafana REST API for multi-environment config management
• Cut trade log extraction latency by 95% (60s → 3s) and eliminated legacy licensing costs by delivering a greenfield FIX log viewer (React, TypeScript, TanStack Table) integrated with Spring Boot APIs

May 2025 – Aug 2025
Singapore

Upside Robotics
Software Engineer Intern
• Enabled ingestion and processing of 5M+ sensor records per run across a live fleet of 10 robots (scalable to 20+) by architecting a serverless AWS pipeline (Lambda, S3, DynamoDB) deployed via IaC (Serverless Framework)
• Reduced manual field diagnostics by 30% by developing a real-time operations dashboard (Next.js, TypeScript, Mapbox) with live telemetry and geospatial tracking
• Ensured reliable data delivery over intermittent LTE networks by collaborating with hardware engineers to implement resilient firmware logic (batching + offline caching)

Aug 2024 – Mar 2025
Waterloo

CUCU Covers
Software Developer Intern
• Improved marketing efficiency by 40% by developing a Slack analytics bot that aggregated Shopify, Instagram, TikTok, and Sendlane data into MongoDB and delivered automated KPI reports (Python/Flask)
• Increased backend throughput 2× by automating order parsing and image processing in a Python pipeline, fully deployed via CI/CD on Vercel
• Reduced operational overhead by 30% by refactoring legacy modules, enabling launch of two new product lines (Device-skins, Wall Art)
• Decreased production incidents by 55% by debugging schema mismatches, API failures, and timestamp errors in production systems

Jan 2024 – Aug 2024
Toronto

PROJECTS

GUI Murphy (TikTok TechJam 2025)
ML Engineer / Team Lead
Developed a multi-stage UI identification and consistency validation pipeline leveraging OmniParser, CLIP, and MLLMs
• Finalist, Top 4% (12 / 300 projects)

Aug 2025 – Sept 2025
Singapore

Smart Storybook, (ETHGlobal SF 2024)
Backend Engineer
A Web3 platform enabling creators to monetize intellectual property through NFTs
• Built a FastAPI backend integrating Pinata's IPFS API for decentralized asset storage
• Won 2nd place, Best AI Application

Oct 2024 – Oct 2024
San Francisco

SKILLS

- Programming Languages:** Python, Java, JavaScript/TypeScript, C, SQL, PHP
- Databases:** MongoDB, MySQL, PostgreSQL, ChromaDB, DynamoDB
- Frameworks & Libraries:** Spring Boot, FastAPI, Flask, React, Next.js, Streamlit, Serverless Framework, Laravel
- DevOps & Tools:** Git, GitHub Actions, Ansible, Redis, ActiveMQ, Hadoop (HDFS & MapReduce), OAuth, Jinja2, pre-commit hooks, Vercel
- Cloud & Infrastructure:** AWS (EC2, S3, Lambda, DynamoDB, Cognito, CloudWatch)